

Basic Characteristics and Characterization of Solar Cells

Solar cells convert power of sunlight into electric power. As an introduction, therefore, Chapter 1 is devoted to a brief characterization of sunlight and basic electric parameters of solar cells. The power of sun is given in terms of the solar constant, the power spectrum and power losses in earth atmosphere expressed by the so-called air mass. The basic characteristics of a solar cell are the short-circuit current (I_{SC}), the open-circuit voltage (V_{OC}), the fill factor (FF) and the solar energy conversion efficiency (η). The influence of both the diode saturation current density and of I_{SC} on V_{OC} , FF and η is analyzed for ideal solar cells. The importance of concentrated sunlight for increasing η is shown. Tolerable series and parallel resistances are introduced as an evaluation criterion for resistive losses in real solar cells. The influence of the series resistance (R_s) and parallel resistance (R_p) on I_{SC} , V_{OC} , FF and η is investigated. The specific role of R_s and R_p is discussed in detail for the dependence of η on I_{SC} . Concepts are described for measuring the basic characteristics of solar cells and their dependencies on light intensity, temperature and light spectra. Attention is paid to principle work with various kinds of load resistances, to the function of a pyranometer, of a sun simulator and to the measurement of the quantum efficiency of solar cells.

1.1 Solar Radiation and Two Fundamental Functions of a Solar Cell

The sun is a hot sphere radiating energy in form of light or photons into space. The absolute temperature (T) of the outer photo sphere of the sun

(T of the sun, T_S) is about 5800 K. The radius of the sun (R_{sun}) is $6.96 \cdot 10^5$ km. The power of thermal radiation and therefore the total power emitted by the sun ($P_{\text{sun}}^{\text{total}}$) can be calculated by using the Stefan–Boltzmann equation and the surface area of the sun.

$$P_{\text{sun}}^{\text{total}} = \sigma_S \cdot T_S^4 \cdot 4\pi \cdot R_{\text{sun}}^2 \quad (1.1)$$

The Stefan–Boltzmann constant (σ_S) is $5.67 \cdot 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$.

The sun's power received on earth (P_e) is proportional to the cross-section of the earth and to the reciprocal area of a sphere with the radius equal to one astronomical unit (AU), the distance between the sun and the earth. This is shown schematically in Figure 1.1. The shortest and longest distances between the sun and the earth, the so-called perihelion and aphelion, are equal to 1.47 and $1.52 \cdot 10^8$ km, respectively. The radius of the earth (R_e) is about 6400 km.

$$P_e = P_{\text{sun}}^{\text{total}} \cdot \frac{\pi \cdot R_e^2}{4\pi \cdot (\text{AU})^2} \quad (1.2)$$

The solar constant (J_s) is defined as the power of the sun (P_{sun}) received on earth over 1 m^2 .

$$J_s \equiv \frac{P_{\text{sun}}^{\text{total}}}{4\pi \cdot (\text{AU})^2} \quad (1.3)$$

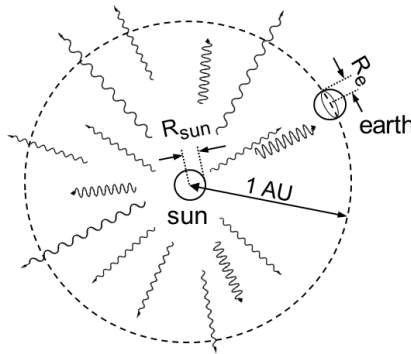


Figure 1.1. Sun emitting photons and earth receiving a proportion of photons emitted by the sun. The radius of the sun, the radius of earth and the distance between sun and earth are denoted by R_{sun} , R_e and AU, respectively.

In reality, the solar constant is not a constant since the distance between the earth and the sun and the temperature distribution at the surface of the sun are not constant. A solar constant of 1356 W/m^2 will be taken into account in the following.

A photon is the smallest portion or quantum of light, the energy of which is proportional to the frequency of the light (ν). The factor between the energy of a photon (photon energy, E_{ph}) and the frequency is called the Planck constant ($h = 6.626 \cdot 10^{-34} \text{ Js}$).

$$E_{\text{ph}} = h \cdot \nu \quad (1.4)$$

The power spectrum of the sun, i.e. the dependence of power emitted within an interval of photon energies ($E_{\text{ph}}, E_{\text{ph}} + dE_{\text{ph}}$), can be approximated by the radiation of a blackbody with the temperature T_S which is given by the Planck equation.

$$\frac{dJ_S(E_{\text{ph}})}{dE_{\text{ph}}} = \frac{8\pi}{h^3 c^3} \cdot \frac{E_{\text{ph}}^3}{\exp\left(\frac{E_{\text{ph}}}{k_B \cdot T_S}\right) - 1} \quad (1.5)$$

The velocity of light (c) is equal to $2.99 \cdot 10^8 \text{ m/s}$ and the Boltzmann constant (k_B) is given as $1.38 \cdot 10^{-23} \text{ J/K}$. The spectrum of blackbody radiation for 5800 K is shown in Figure 1.2.

Spectra are usually measured in units of wavelength (λ). The λ of light is proportional to the reciprocal frequency while the proportionality factor is the velocity of light.

$$\lambda = \frac{c}{\nu} \quad (1.6)$$

The E_{ph} is given in units of eV, which means that the energy is divided by the elementary charge ($q = 1.6 \cdot 10^{-19} \text{ As}$). Therefore, the product of the photon energy and the wavelength is given by

$$E_{\text{ph}} \cdot \lambda = \frac{h \cdot c}{q} = 1240 \text{ eV} \cdot \text{nm} \quad (1.7)$$

The λ can be easily transformed into the E_{ph} by using Equation (1.7). The E_{ph} and the λ in the maximum of the corresponding spectra of blackbody radiation are obtained in tasks T1.3 and T1.4 (see end of this chapter), respectively, for a blackbody with T of 5800 K .

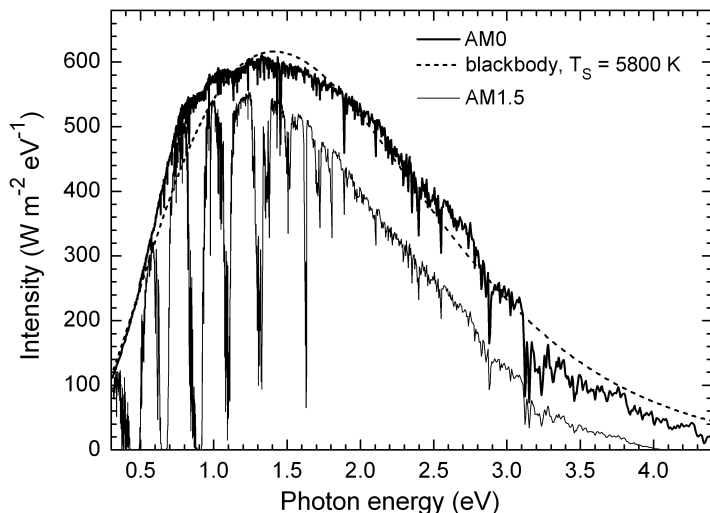


Figure 1.2. Spectrum of sunlight (sun spectrum) outside the earth's atmosphere (air mass (AM) 0, thick solid line), on earth for a zenith angle of 48.2° (AM1.5, thin solid line) and of a blackbody with a temperature of 5800 K (dashed line).

The power received over a certain area on earth depends on geographical location, on the rotation of the earth (day–night cycle), on the inclination angle of the earth's axis (sun in summer and winter) and on the given distance between sun and earth. Further, light is scattered and absorbed by molecules and particles in the earth's atmosphere. The average power of sunlight received on earth, for example, is reduced by a factor of π as a result of the day–night cycle.

The optical path of sunlight through the earth's atmosphere normalized to the thickness of the earth's atmosphere defines the so-called air mass (AM). The power of the sun corresponds at zenith to AM1 and at a zenith angle of 48.2° to AM1.5. It is agreed worldwide that P_{sun} is equal to 1 kW/m^2 at AM1.5G (IEC, 2008) (G denotes global radiation including direct light and scattered light, see also the AM1.5 spectrum (the sun spectrum can be downloaded, for example from (RReDc, 2013)) in Figure 1.2; where direct radiation is less than the global radiation by a factor of about 1.1). The average power of the sun on earth ($\langle P_{\text{sun}} \rangle$) is, for example, 0.12 kW/m^2 in Germany and between 0.2 and 0.27 kW/m^2 in most regions in the so-called sunbelt (i.e. those between about the -30 and $+30^\circ$ of latitude).

A solar cell converts P_{sun} into electric power (P), i.e. the product of electric current (I) and electric potential or voltage (U).

$$P = I \cdot U \quad (1.8)$$

With respect to Equation (1.8), the two fundamental functions of a solar cell are (i) the photocurrent generation and (ii) the generation of a photovoltage.

Photocurrent generation means the creation of mobile photogenerated charge carriers by absorbing light and their collection at external contacts. A potential difference between external contacts is necessary for the collection of photogenerated charge carriers. It is not important for the measurement of a photocurrent whether the potential difference between external contacts originates from an external or internal voltage source. Charge separation at a charge-selective contact inside the solar cell is the origin for an internal voltage source and leads to the generation of a photovoltage.

The current or photocurrent is often normalized to the area of the solar cell under consideration. Current and current density will not be distinguished in the following since their meaning follows directly from the unit used (A for current and A/cm² for current density) or from the context.

The electric power of solar cells and photovoltaic (PV) modules is on the order of 1 mW to 300 W. PV power plants can be installed for the kW–MW range, and even higher. The extreme scalability of solar cells and PV power plants over many orders of magnitude makes the application of PV solar energy conversion very flexible. This is unique in comparison to any other technology of electricity production.

The installed power of a PV power plant (P_{inst}) is defined as the maximum power which can be given by the power plant at P_{sun} (AM1.5) during the first year after installation. P_{inst} is proportional to the area covered by the PV modules (A) and their solar energy conversion efficiency (η , Equation (1.15)). P_{inst} is given in W_p (Watt peak).

$$P_{\text{inst}} = A \cdot \eta \cdot P_{\text{sun}} \text{ (AM1.5)} \quad (1.9)$$

The electric energy (W_{el}) which can be produced with a PV power plant is the product of A , η , $\langle P_{\text{sun}} \rangle$ and the time period of operation (t).

$$W_{\text{el}} = A \cdot \eta \cdot \langle P_{\text{sun}} \rangle \cdot t \quad (1.10)$$

PV power plants can be described by the ratio between W_{el} and P_{inst} .

$$\frac{W_{\text{el}}}{P_{\text{inst}}} = \frac{\langle P_{\text{sun}} \rangle}{P_{\text{sun}}(\text{AM1.5})} \cdot t = \left[\frac{\text{kWh}}{\text{kW}_p} \right] \quad (1.11)$$

The η of PV modules decreases during operation time at a so-called degradation rate (k_{deg}) of between about 0.3% and 1.5% per year (Jordan *et al.*, 2016). The degradation time of a PV module (Δt_{η}) can be defined as the time after which η reduced to a certain value (see also task T1.2).

The energy payback time of a PV module (t_{EPB}) is defined as the operation time of the PV module resulting in W_{el} equal to the energy which was treated for the production of the corresponding PV module. Low values of k_{deg} and t_{EPB} are important for high energy pay back and therefore for sustainable PV solar energy conversion (see also task T1.3). Changing environmental parameters, such as T and humidity, cause stress in materials and combinations of materials. Therefore, degradation rates depend on the stability of materials and combinations of materials under stress induced, for example, by different thermal expansion coefficients of materials.

1.2 Basic Characteristics of a Solar Cell

An illuminated solar cell can provide a certain photovoltage at a given photocurrent. A combination of values of photocurrent and photovoltage at which a solar cell can be operated is called a working point. A particular working point of a solar cell is fixed with a load resistance (R_L) due to the Ohm's law.

$$R_L = \frac{U}{I} \quad (1.12)$$

In terms of Ohm's law, the photovoltage is very low at very low R_L , and the photocurrent is very low at very high R_L . The short- and open-circuit operation conditions of a solar cell are defined as a R_L which is equal to zero or which is infinitely high, respectively. The values of the photocurrent and of the photovoltage at short- and open-circuit conditions are called short-circuit current (I_{SC}) and open-circuit voltage (V_{OC}), respectively. The electric power is equal to zero at short- and open-circuit operation of a solar cell.

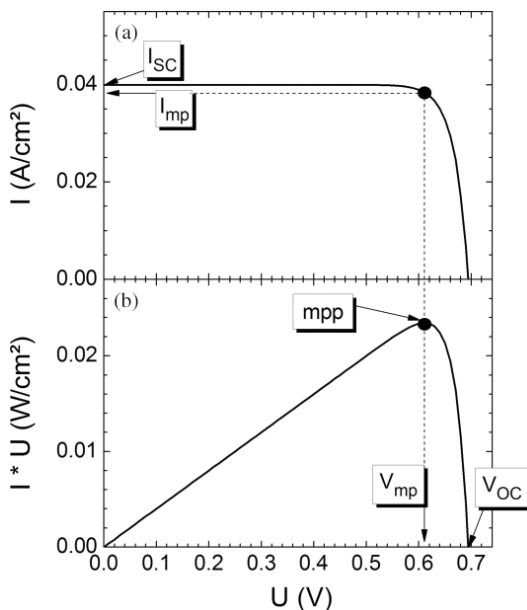


Figure 1.3. Example of a current–voltage characteristic (a) and of the corresponding power–voltage characteristic (b) of a solar cell under illumination. The short-circuit current density, the open-circuit voltage, the maximum power point and the voltage and current density at the maximum power point are denoted by I_{SC} , V_{OC} , mpp, V_{mp} and I_{mp} , respectively.

A current–voltage characteristic (I – V characteristic) of a solar cell is a plot of all possible working points in a considered range. Figure 1.3 shows schematically the I – V characteristic of a solar cell under illumination.

There is one combination of current and voltage at which the power of the solar cell has its maximum (I_{mp} and V_{mp} , respectively). This point on the I – V characteristic of an illuminated solar cell is called the maximum power point (mpp).

$$P_{mpp} = I_{mp} \cdot U_{mp} \quad (1.13)$$

The values of I_{SC} and V_{OC} can be measured easily. Therefore, it is convenient to characterize the maximum power of a solar cell with I_{SC} , V_{OC} and an additional parameter instead of characterizing the solar cell with I_{mp} and U_{mp} . The additional parameter sets the product of I_{mp} and U_{mp} in relation to the product of I_{SC} and V_{OC} . This parameter describes the

amount by which the I_{SC} – V_{OC} rectangle is filled by the I_{mp} – U_{mp} rectangle and is therefore called the fill factor (FF).

$$FF = \frac{I_{mp} \cdot U_{mp}}{I_{SC} \cdot V_{OC}} \quad (1.14)$$

The η of a solar cell is defined as the ratio between the power extracted at the mpp of the solar cell and the power of the sunlight at which the solar cell is illuminated (P_{sun}).

$$\eta = FF \frac{I_{SC} \cdot V_{OC}}{P_{sun}} \quad (1.15)$$

The solar energy conversion efficiency is a decisive parameter for costs and sustainability of PV energy production. The higher the value of η , the lower the amount of material and area needed for a PV power plant with a given P_{inst} . A lot of effort has been directed in the past and will continue to be in the future (i) to the investigation of fundamental limitations of η for different types of solar cells, (ii) to materials science for finding out suitable materials and materials combinations for high values of η and (iii) to the development of technologies allowing the realization of maximum values of η in mass production and with a minimum of resources.

1.3 Ideal Solar Cell Under Illumination

1.3.1 Diode equation of the ideal solar cell

The two fundamental functions of a solar cell are expressed by two different elements in the equivalent circuit of the ideal solar cell (Figure 1.4). The first element is a current source driven by illumination, i.e. the photocurrent generator. The second element has to fulfill the condition of charge separation, i.e. the current can pass through the element in one

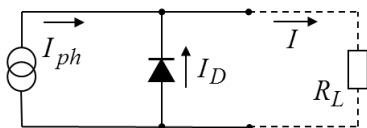


Figure 1.4. Equivalent circuit of an ideal solar cell containing a photocurrent generator and a diode for charge separation and connected to a R_L .

direction but not in the other. This is the characteristic property of a diode, the second element in the equivalent circuit of an ideal solar cell.

A current across a diode (I_D) is described by the diode equation

$$I_D = I_0 \cdot \left[\exp\left(\frac{q \cdot U}{k_B \cdot T}\right) - 1 \right] \quad (1.16)$$

The diode equation contains the proportionality factor which is called the diode saturation current or diode saturation current density (I_0). The diode saturation current is a specific characteristic of each solar cell depending on the absorber and contact materials as well on their geometry. The diode saturation current is multiplied with an exponential of the voltage while the voltage is multiplied with the q and divided by the product of k_B and T . The diode current increases exponentially with increasing positive voltage (forward direction or forward bias). At zero voltage, the diode current should be equal to zero. It is for this reason that 1 has to be subtracted from the exponential term in the diode equation. At negative voltage, the diode current decreases to the very low negative I_0 (reverse direction or reverse bias).

Any charge-selective contact can be described by a diode equation. The diode equation will be derived for the pn-junction in Chapter 4 and for the ideal solar cell in the Shockley–Queisser limit in Chapter 6.

The resistance of an ideal diode is extremely low in forward direction and very large in reverse direction. The photocurrent flows through the R_L , i.e. the photocurrent is not shunted by the diode. Therefore the photocurrent is a reverse current and has to be subtracted from the diode current. This results in the diode equation of the ideal solar cell.

$$I = I_0 \cdot \left[\exp\left(\frac{q \cdot U}{k_B \cdot T}\right) - 1 \right] - I_{ph} \quad (1.17)$$

Figure 1.5 shows I – V characteristics of an ideal solar cell with $I_0 = 10^{-13} \text{ A/cm}^2$ in the dark and under illumination ($I_{ph} = 0.04 \text{ A/cm}^2$).

Equation (1.17) is used for the analysis of I – V characteristics of illuminated solar cells by setting I_{ph} equal to I_{SC} . Typical values of the I_{SC} are of the order of tens of mA/cm^2 at AM1.5.

The I – V characteristics of solar cells under illumination are often plotted in the first quadrant for convenience. For this purpose, the

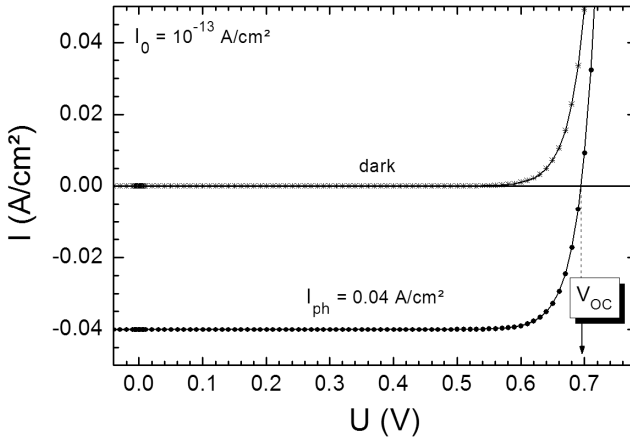


Figure 1.5. Current–voltage characteristics of an ideal solar cell with $I_0 = 10^{-13} \text{ A/cm}^2$ in the dark (stars) and under illumination at $I_{ph} = 0.04 \text{ A/cm}^2$ (circles).

negative current is plotted in the I – V characteristic of a solar cell. As a convention, the negative I_{ph} corresponds to I_{SC} .

$$-I = I_0 \cdot \left[\exp\left(\frac{q \cdot U}{k_B \cdot T}\right) - 1 \right] - I_{SC} \quad (1.18)$$

1.3.2 Open-circuit voltage and diode saturation current

The V_{OC} of an ideal solar cell can be obtained from Equation (1.18) by setting the current to zero.

$$V_{OC} = \frac{k_B \cdot T}{q} \cdot \ln\left(\frac{I_{SC}}{I_0} + 1\right) \approx \frac{k_B \cdot T}{q} \cdot \ln\left(\frac{I_{SC}}{I_0}\right) \quad (1.19)$$

The value of I_0 can change over many orders of magnitude depending on the materials and on interfaces between materials forming the diode. The importance of I_0 for the performance of a solar cell follows from the limitation of V_{OC} by I_0 . The minimization of I_0 is crucial for the optimization of the performance of solar cells in general.

The value of V_{OC} increases with increasing I_{SC} . The dependence of V_{OC} on I_{SC} is shown in Figure 1.6(a) for three different values of I_0 .

It is useful to remember that $k_B \cdot T/q$ amounts to 0.026 V at room temperature and that the natural logarithm of 10 is equal to 2.3. Therefore,

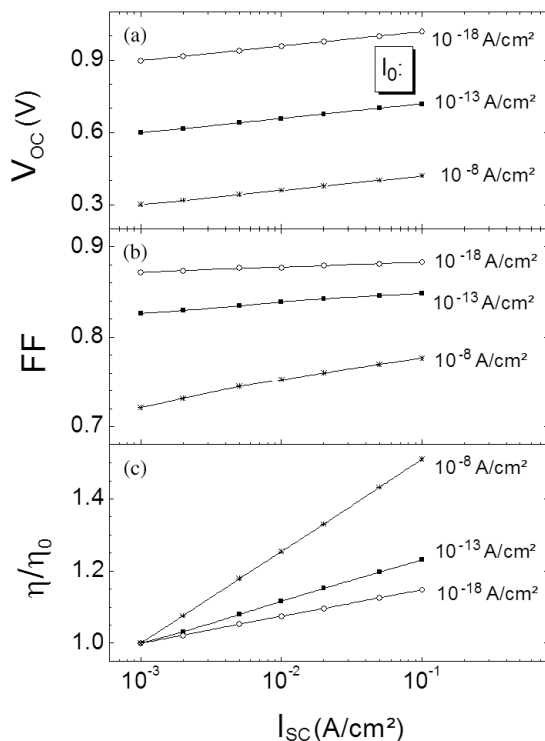


Figure 1.6. Dependence of V_{OC} (a), FF (b) and the normalized η (c) on I_{SC} for an ideal solar cell with $I_0 = 10^{-18}$, 10^{-13} and 10^{-8} A/cm² (squares, circles and stars, respectively).

for an ideal solar cell, V_{OC} increases at room temperature by 60 mV if I_{SC} increases by one order of magnitude.

$$\left[\frac{\Delta V_{OC}}{\text{decade of } I_{SC}} \right]_{300\text{ K}} = 60 \text{ mV per decade} \quad (1.20)$$

The order of magnitude of I_0 can be estimated for crystalline silicon (c-Si) and gallium arsenide (GaAs) record solar cells if assuming them to be ideal.

$$I_0 = \frac{I_{SC}}{\exp\left(\frac{qV_{OC}}{k_B \cdot T}\right)} \quad (1.21)$$

The values of I_{SC} and V_{OC} are 41.8 mA/cm² and 0.74 V, respectively, for a c-Si record solar cell and 29.68 mA/cm² and 1.122 V, respectively,

for a GaAs record solar cell (Green *et al.*, 2016). With respect to Equation (1.18) for I_0 one gets

$$I_0(\text{c-Si}) \approx 2 \cdot 10^{-14} \text{ A/cm}^2 \quad (1.22')$$

$$I_0(\text{GaAs}) \approx 6 \cdot 10^{-21} \text{ A/cm}^2 \quad (1.22'')$$

Therefore, I_0 of high-performance c-Si solar cells has to be about 10^{-13} A/cm^2 . In comparison to c-Si solar cells, I_0 can be reduced by about 6 orders of magnitude for GaAs solar cells. The reason for this difference is explored in Chapter 2 and derived in Chapter 4.

1.3.3 Dependence of the fill factor on the short-circuit current density

The FF (see Equation (1.14)) can be calculated by simulating Equation (1.18) and finding I_{mp} and U_{mp} . As an example, the FF is plotted as a function of I_{SC} for $I_0 = 10^{-18}$, 10^{-13} and 10^{-9} A/cm^2 in Figure 1.6(b). At fixed I_{SC} the FF is higher for the lower I_0 . The FF increases with increasing I_{SC} at fixed I_0 . The increase of FF with increasing I_{SC} is weak for low values of I_0 . For example, at $I_0 = 10^{-18} \text{ A/cm}^2$ the FF changes roughly between 0.87 and 0.88 for I_{SC} between 1 and 100 mA/cm^2 . On the other hand, the increase of FF with increasing I_{SC} is stronger for lower values of I_0 . For example, at $I_0 = 10^{-8} \text{ A/cm}^2$ the FF changes roughly between 0.72 and 0.78 for I_{SC} between 1 and 100 mA/cm^2 .

Due to the influence of I_0 , the FFs of solar cells made from similar materials and materials combinations are practically identical at a given I_{SC} if a similar architecture and fabrication technology were used for all compared solar cells. Therefore, besides the relatively weak change of the FF with I_{SC} , FF obtains specific values depending on the type of a solar cell and its production process. For example, FF is lower for thin-film solar cells (see Chapter 9) than for c-Si solar cells (see Chapter 7).

1.3.4 Dependence of the efficiency of an ideal solar cell on the light intensity

The solar energy conversion efficiency depends on P_{sun} since FF, I_{SC} and V_{OC} depend on light intensity. The ratio of two solar energy conversion efficiencies (η/η_0), where the index 0 is related to the lower value of P_{sun} ,

are considered in the following:

$$\frac{\eta}{\eta_0} = \frac{FF}{FF_0} \cdot \frac{I_{SC}}{I_{SC,0}} \cdot \frac{V_{OC}}{V_{OC,0}} \cdot \frac{P_{\text{sun},0}}{P_{\text{sun}}} \quad (1.23)$$

Equation (1.23) can be simplified if assuming that I_{SC} increases linearly with increasing light intensity, i.e. I_{SC} is proportional to P_{sun} . With respect to Equation (1.19), therefore, one gets

$$\frac{\eta}{\eta_0} = \frac{FF}{FF_0} \cdot \frac{T}{T_0} \cdot \frac{\ln \frac{I_{SC}}{I_0}}{\ln \frac{I_{SC,0}}{I_{0,0}}} \quad (1.24)$$

Equation (1.24) can be further simplified if heating due to increased P_{sun} is neglected and if a constant I_0 is assumed. Of course, one has to be careful with both of these assumptions since the T of solar cells generally increases with increasing P_{sun} and since I_0 can depend on P_{sun} . After a transformation, the following equation is obtained:

$$\frac{\eta}{\eta_0} = \frac{FF}{FF_0} \cdot \left[1 + \frac{X}{\ln \frac{I_{SC,0}}{I_0}} \right] \quad (1.25)$$

X is the so-called concentration factor.

$$X = \frac{P_{\text{sun}}}{P_{\text{sun},0}} \quad (1.26)$$

The ratio of the FFs is larger than 1, as determined in Section 1.3.3, and X and the natural logarithm of the ratio of $I_{SC,0}$ and I_0 are positive. Therefore, η of an ideal solar cell is larger at the higher light intensity. This is the reason why the absolute largest solar energy conversion efficiencies are reached under concentrated sunlight.

The solar energy conversion efficiency of an ideal solar cell normalized to η at $I_{SC} = 1 \text{ mA/cm}^2$ is plotted in Figure 1.6(c) as a function of I_{SC} for $I_0 = 10^{-18}$, 10^{-13} and 10^{-9} A/cm^2 . If I_{SC} increases by two orders of magnitude, the relative increases of η is 13, 20 and 50% for $I_0 = 10^{-18}$, 10^{-13} and 10^{-9} A/cm^2 , respectively. However, it has to be taken into account that the assumptions of (i) linear I_{SC} response, (ii) absence of additional heating and (iii) independence of I_0 on P_{sun} may not be fulfilled for real solar cells so that Equation (1.25) can no longer be applied.

1.4 Real Solar Cells: Consideration of Series and Parallel Resistances

1.4.1 Resistive losses and tolerable series and parallel resistances

In reality, any solar cell has losses due to ohmic resistances. For example, contact resistances or material resistances are responsible for series resistances and local shunts or other imperfections cause parallel resistances in a solar cell. One common series resistance (R_s) and one common parallel resistance (R_p), which is also called shunt resistance, are considered in the equivalent circuit of a real solar cell (Figure 1.7).

The current flowing through the R_s is equal to the current flowing through R_L . Therefore, the potential across the R_L is reduced by the voltage drop across the R_s . Further, the current flowing through the R_L is reduced by the current flowing through the R_p . The voltage drop across R_s and the current shunted by R_p are taken into account in the following diode equation:

$$-I = I_0 \cdot \left[\exp\left(\frac{q \cdot (U - I \cdot R_s)}{k_B \cdot T}\right) - 1 \right] + \frac{U - I \cdot R_s}{R_p} - I_{SC} \quad (1.27)$$

The voltage drop across R_s is assumed to be significantly smaller than U_{mp} . Then, the power losses due to resistive heating resistive power losses of R_s and R_p are given by

$$P_{\text{loss}}(R_s) = R_s \cdot I_{mp}^2 \quad (1.28')$$

$$P_{\text{loss}}(R_p) \approx \frac{U_{mp}^2}{R_p} \quad (1.28'')$$

It is useful at this juncture to introduce so-called tolerable series and parallel resistances ($R_{s,\text{tol}}$ and $R_{p,\text{tol}}$, respectively) in accordance to the power losses that can be tolerated in comparison to the power in the mpp. It makes

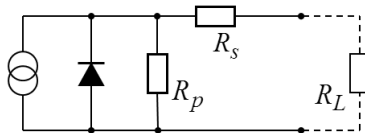


Figure 1.7. Equivalent circuit of a real solar cell containing a photocurrent generator, an ideal diode, a shunt resistance (R_p) and a series resistance (R_s) and being connected with a load resistance (R_L).

sense to tolerate a power loss caused by resistive heating of about 1% of the power in the mpp of a solar cell with very high η . The R_L in the mpp is of the same order of magnitude as the ratio of V_{OC} and I_{SC} (R_L^*). As an approximation rule, one can write the following conditions for $R_{s,tol}$ and $R_{p,tol}$:

$$R_{s,tol} < \frac{1}{100} \cdot \frac{V_{OC}}{I_{SC}} = \frac{R_L^*}{100} \quad (1.29')$$

$$R_{p,tol} > 100 \cdot \frac{V_{OC}}{I_{SC}} = 100 \cdot R_L^* \quad (1.29'')$$

Equations (1.29) describe rather harsh conditions. For example, a solar cell based on a c-Si wafer (for details about c-Si solar cells, see in Chapter 7) with an area of about 100 cm^2 has an I_{SC} of about 3–4 A and a V_{OC} of about 0.6–0.7 V. This means that the overall R_s or R_p of this solar cell should be less than about $2 \text{ m}\Omega$ or larger than 20Ω , respectively.

A major source for R_s in c-Si solar cells are the bus bars collecting the charge carriers from the whole solar cell. The bus bars are mainly based on silver. The resistance of a bus bar can be calculated if the specific conductivity (σ , $6.2 \times 10^7 \text{ S/m}$ for silver), the length (L), the width (B) and the height (H) of the bus bar are known.

$$R = \frac{L}{\sigma \cdot B \cdot H} \quad (1.30)$$

The resistance is about $3 \text{ m}\Omega$ for a typical bus bar with a length of 10 cm, a width of 5 mm and a height of 0.1 mm which is larger than the $2 \text{ m}\Omega$ demanded. Crystalline silicon solar cells have usually several bus bars depending on the area of the solar cell and on the resistance of the emitter (see Chapter 7).

Current and voltage can be related to area fractions of a solar cell, to solar cells connected in PV modules, to PV modules connected in strings or to strings of PV modules connected in large PV power plants. The conditions for the area of the cross section of bus bars in c-Si solar cells or of copper cables connecting strings of PV modules are equivalent and can be obtained by using Equations (1.29') and (1.30). For this purpose, the series and parallel connection of solar cells has to be taken into account. In a series connection, the current remains constant whereas the individual

values of the voltage are added.

$$U = \sum_i U_i \quad (1.31')$$

$$I = I_i \quad (1.31'')$$

For parallel connection, the voltage remains unchanged and the individual values of the current are added.

$$U = U_i \quad (1.32')$$

$$I = \sum_i I_i \quad (1.32'')$$

Series connection is favored if an increase of the area of the cross section of connecting cables and therefore of weight or amount of material is unwanted. Binning of solar cells or PV modules in narrow ranges is very important for series connection since the solar cell or the PV module with the lowest photocurrent limits the photocurrent of all other connected solar cells or PV modules. As a consequence the area fraction in an isolated solar cell, the solar cell or the PV module in a string of PV modules with the lowest η will limit the efficiency of the whole system where they are part, i.e. of the complete solar cell, of the PV module or of the PV power plant. This is a reason, for example, why only solar cells with an area of at least 1 cm^2 are considered for world records (AM1.5) and why η decreases with increasing area of solar cells, PV modules or PV power plants.

1.4.2 Influence of series and parallel resistances on the basic characteristics of a solar cell

The R_s and R_p can heavily influence the I - V characteristics of a solar cell. The influence of R_s and R_p on I - V characteristics will be analyzed separately in the following for a solar cell with fixed I_0 (10^{-13} A/cm^2) and fixed I_{SC} (0.04 A/cm^2), which is close to the record c-Si solar cells illuminated at P_{sun} (AM1.5). Under these conditions, V_{OC} is close to 0.7 V and R_L^* is about $17 \Omega \text{ cm}^2$ in case of the ideal solar cell (R_{Lmp}^{ideal}). As remark, the I_{SC} in Equation (1.27) and the short-circuit current obtained at zero potential (I_{SC}^* , see also Figure 1.8(a)) can be rather different depending on R_s and thus have to be distinguished.

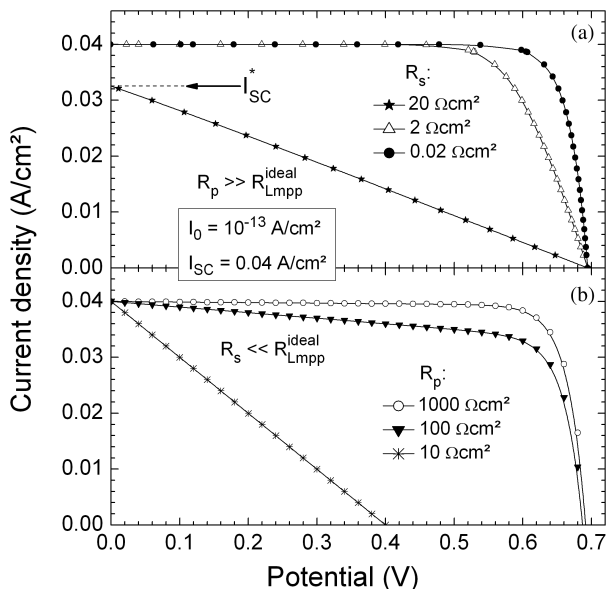


Figure 1.8. I – V characteristics for a solar cell with negligible R_p (a) or R_s (b) for different values of R_s (a) and R_p (b) and $I_0 = 10^{-13} A/cm^2$ and $I_{SC} = 0.04 A/cm^2$.

Figure 1.8(a) shows the I – V characteristics for solar cells for which R_p can be neglected and for which the R_s are about three or one orders of magnitude lower than R_{Lmpp}^{ideal} or very close to R_{Lmpp}^{ideal} . The values of I_{SC} and V_{OC} are practically identical for the I – V characteristics with $R_s = 0.02 \Omega cm^2$ and $R_s = 2 \Omega cm^2$, while FF decreases significantly for the I – V characteristic with $R_s = 2 \Omega cm^2$. The value of V_{OC} remains constant with further increase of R_s since R_s has no influence if there is no current flowing. However, I_{SC}^* decreases to about $0.032 A/cm^2$, which is less than I_{SC} , and the I – V characteristic becomes a straight line for $R_s = 20 \Omega cm^2$. The FF is 0.25 for a linear I – V characteristic. An FF of 0.25 corresponds to the minimum possible FF of a solar cell which can be described with an equivalent circuit as shown in Figure 1.7. High values of R_s limit I_{SC} and FF.

Figure 1.8(b) shows the I – V characteristics for solar cells in which R_s can be neglected and for which the R_p are about two or one orders of magnitude larger than R_{Lmpp}^{ideal} or very close to R_{Lmpp}^{ideal} . The values of I_{SC} are identical for all R_p since the shunt resistance can be neglected under the I_{SC} condition corresponding to an infinitely low R_L . The value

of V_{OC} is a little bit lower for the solar cell with $R_p = 100 \Omega\text{cm}^2$ than for the solar cell with $R_p = 1000 \Omega\text{cm}^2$ and decreases to 0.4 V for the solar cell with $R_p = 10 \Omega\text{cm}^2$. The I - V characteristic becomes a straight line for $R_p = 10 \Omega\text{cm}^2$, i.e. FF is 0.25. Low values of R_p limit V_{OC} and FF.

The dependencies of V_{OC} , I_{SC}^* , FF and of the normalized η on R_s or R_p are presented in Figures 1.9(a)–1.9(d), respectively, for the same values of I_0 and I_{SC} as in Figure 1.8.

The values of V_{OC} or I_{SC} are practically constant for $R_p > R_{Lmp}^{ideal}$ or for $R_s < R_{Lmp}^{ideal}$, respectively, and decrease linearly with decreasing R_p ($R_p < R_{Lmp}^{ideal}$) or increasing R_s ($R_s > R_{Lmp}^{ideal}$), respectively. The FFs decrease with decreasing R_p ($R_p \leq 100 \cdot R_{Lmp}^{ideal}$) or increasing R_s ($R_s \geq R_{Lmp}^{ideal}/100$), respectively, and remain practically constant at 0.25 for $R_p < R_{Lmp}^{ideal}$ or

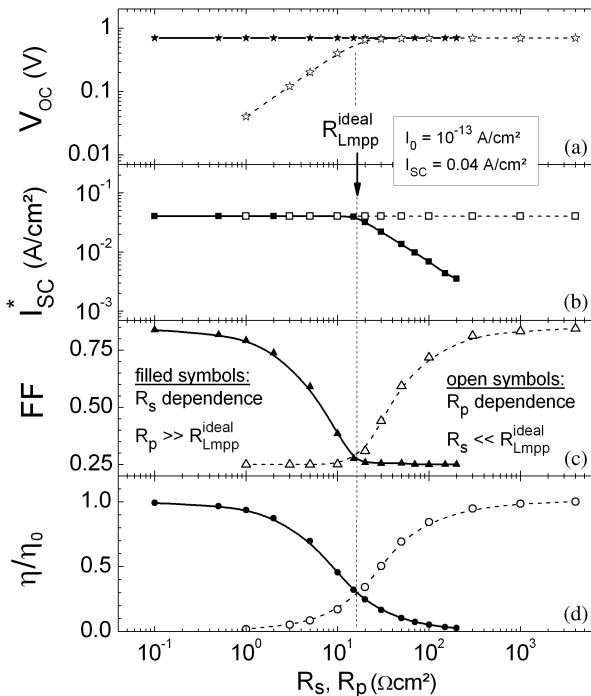


Figure 1.9. Dependence of the V_{OC} (a), the I_{SC}^* density (b), the FF (c) and the normalized η (d) on the R_s (filled symbols) and R_p (open symbols) for solar cells with $I_0 = 10^{-13} \text{ A/cm}^2$, $I_{SC} = 0.04 \text{ A/cm}^2$.

for $R_s > R_{Lmp}^{ideal}$, respectively. Therefore, η is limited by FF in the ranges $R_{Lmp}^{ideal} \leq R_p \leq 100 \cdot R_{Lmp}^{ideal}$ and $R_L \geq R_s \geq R_{Lmp}^{ideal}/100$ and by V_{OC} and I_{SC} in the ranges of $R_p < R_{Lmp}^{ideal}$ and $R_s > R_{Lmp}^{ideal}$, respectively.

The qualitative influence of R_p and R_s on the basic characteristics of a solar cell is generally similar to the dependencies shown in Figures 1.8 and 1.9 where R_s and R_p should be scaled with respect to I_0 and I_{SC} .

1.4.3 Influence of series and parallel resistances on the intensity dependence of the basic characteristics of a solar cell

The influence of R_s and R_p on the I - V characteristics of a solar cell can change strongly with changing light intensity. For this reason R_s and R_p have a tremendous influence on the intensity dependence of the basic characteristics of a solar cell. The I_{SC} in the diode equation increases linearly with increasing P_{sun} .

The influence of I_{SC} on the I - V characteristics will be analyzed separately in the following for a solar cell with fixed I_0 (10^{-13} A/cm²) and given values of R_s ($10 \Omega\text{cm}^2$) and R_p ($100 \Omega\text{cm}^2$), where the respective values of R_p ($R_s = 10 \Omega\text{cm}^2$) and R_s ($R_p = 100 \Omega\text{cm}^2$) are neglected (Figures 1.10(a) and 1.10(b), respectively).

As shown in Figure 1.10(a) the FF decreases with increasing I_{SC} for a given R_s . The value of I_{SC}^* is proportional to I_{SC} at values for which $I_{SC} < V_{OC}/R_s$. The FF reaches 0.25 when I_{SC} becomes larger than V_{OC}/R_s . In this region I_{SC}^* becomes proportional to V_{OC} . The FF decreases with decreasing I_{SC} for a given R_p (Figure 1.10(b)). The logarithm of the value of V_{OC} is proportional to I_{SC} at values for which $I_{SC} > V_{OC}/R_p$. The FF reaches 0.25 when I_{SC} becomes lower than V_{OC}/R_p . In this region V_{OC} decreases linearly with decreasing I_{SC} .

The voltage drop across the R_s increases with increasing current, which leads to a decrease of the η . This means that R_s limits the η at high values of I_{SC} . On the other hand, the influence of R_p increases with decreasing I_{SC} , R_p limits the η at low values of I_{SC} . The intensity dependencies of V_{OC} , I_{SC}^* and FF can be summarized as the dependencies of the normalized η on I_{SC} .

Figure 1.11 shows the dependence of the normalized η of solar cells with $I_0 = 10^{-13}$ A/cm² for R_s of 0.01, 1 and $100 \Omega\text{cm}^2$ (a) and for R_p of 1, 100 and $10,000 \Omega\text{cm}^2$ (b). The range of I_{SC} is chosen in such a way

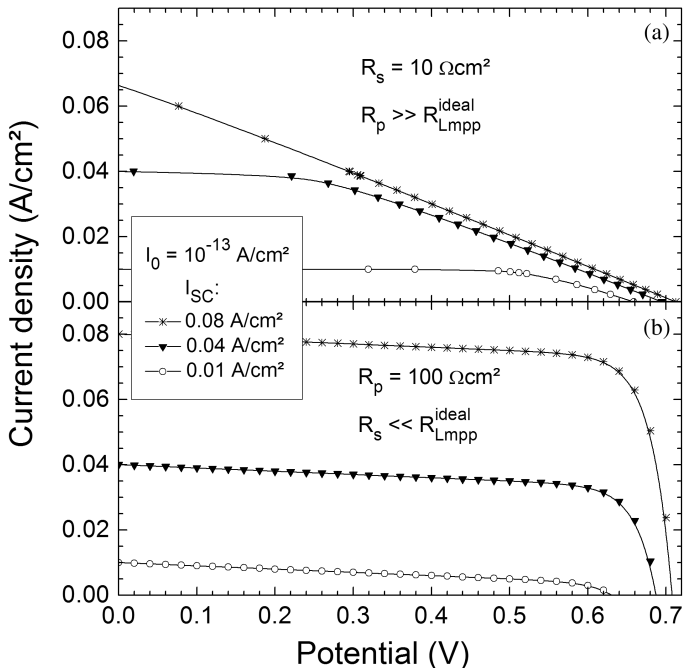


Figure 1.10. I - V characteristics for solar cells with negligible R_p (a) or R_s (b) for $I_0 = 10^{-13} A/cm^2$ and $I_{SC} = 0.08, 0.04$ and $0.01 A/cm^2$ (stars, triangles and circles, respectively) for $R_s = 10 \Omega cm^2$ (a) and $R_p = 100 \Omega cm^2$ (b). Incidentally, I_{SC} corresponds to the value in the diode equation.

that the influence of concentrated sunlight or of weak illumination can be analyzed. The illumination target range of the application of a given type of solar cell has significant technological and economic consequences. Concentration factors of up to 1000 are important for highly efficient concentrator solar cells. At very high concentration factors R_s should be significantly lower than $0.01 \Omega cm^2$.

The R_p of c-Si solar cells fabricated in conventional mass production is only of the order of several hundred Ωcm^2 , which is not favorable for highly efficient operation at reduced light intensity. More energy can be produced over the year with a PV power plant based on solar cells with higher R_p due to reduced losses in the morning and evening hours. For example, high values of R_p can be reached with amorphous silicon solar cells (see Chapter 9).

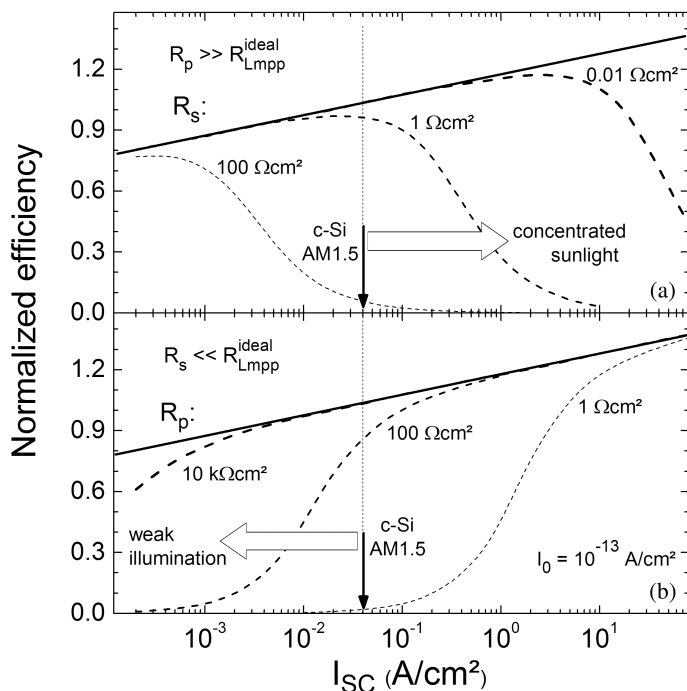


Figure 1.11. Dependencies of the normalized solar energy conversion efficiency on I_{SC} for a solar cell with $I_0 = 10^{-13} \text{ A/cm}^2$ at different values of R_s (a) (R_p neglected) and R_p (b) (R_s neglected). The thin and thick arrows mark I_{SC} of an ideal c-Si solar cell at AM1.5 and the directions towards concentrated sunlight and weak illumination, respectively.

Light intensities reduced by up to about 2 orders of magnitude in comparison to AM1.5 can make sense for low-power indoor applications of solar cells and can be relevant for standby functions of electronic devices. The highest values of R_p can be achieved with dye sensitized solar cells (see Chapter 10).

1.5 Characterization of Solar Cells

1.5.1 I_{SC} – V_{OC} characteristics and ideality factor

The simplest measurements on solar cells are performed with just one multimeter by measuring the V_{OC} as a function of the I_{SC} . Regarding to Equations (1.19) and (1.27), considering a low R_s and taking into account

an ideality factor (n), the following I_{SC} – V_{OC} characteristic is obtained:

$$I_{SC} = I_0 \cdot \left[\exp\left(\frac{q \cdot V_{OC}}{n \cdot k_B \cdot T}\right) - 1 \right] + \frac{V_{OC}}{R_p} \quad (1.33)$$

It has to be taken into account that a high accuracy for current measurement over 4–6 orders of magnitude is required. An intensity control of the light is not needed since I_{SC} and V_{OC} are correlated directly and measured at identical light intensities.

Equation (1.33) has I_0 , n and R_p as parameters which can be determined from the I_{SC} – V_{OC} characteristic (see Figure 1.12). An I_{SC} – V_{OC} characteristic can be separated into two parts that are dominated by the exponential term and by the ohmic term at high and low values of V_{OC} , respectively.

The slope of the exponential term is usually significantly larger than 60 mV per decade, i.e. a higher potential than for an ideal solar cell is needed to increase the current by one order of magnitude. The ratio of the measured slope and 60 mV per decade is called the ideality factor and is equal to one for an ideal solar cell. The ideality factor is usually on the order of 1.1–1.3 for c-Si solar cells. The value of I_0 is also found from the

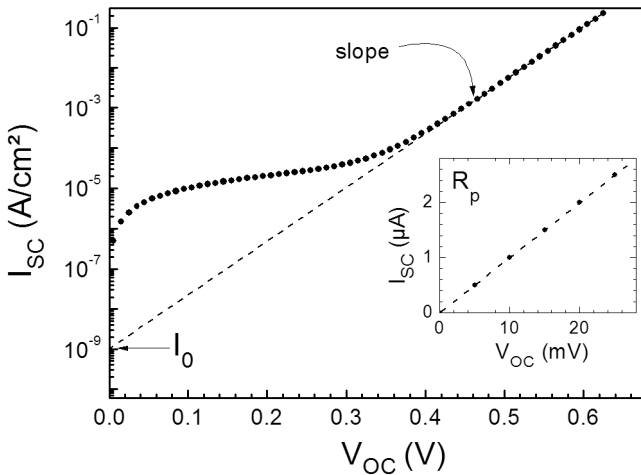


Figure 1.12. Dependence of the I_{SC} on the V_{OC} for a given solar cell in a logarithmic scale and in a linear scale (zoom, inset).

exponential part of the I_{SC} – V_{OC} characteristic. The value of R_p is found from the slope of the ohmic part of the I_{SC} – V_{OC} characteristic.

1.5.2 Temperature-dependent diode saturation current and activation energy

The value of I_0 can be obtained from I_{SC} – V_{OC} characteristics measured at different T . The temperature dependence of I_0 is usually plotted in a so-called Arrhenius plot. Arrhenius plots generally present a T -dependent parameter on a logarithmic scale as a function of T in a reciprocal scale. Figure 1.13 shows an example of a typical Arrhenius plot for I_0 .

Arrhenius plots allow for the extraction of the relevant parameter controlling the T dependence of I_0 . The point is that the T dependence of I_0 usually follows a straight line in the Arrhenius plot, which means that I_0 depends exponentially on the T . The control parameter is the thermal activation energy (E_A) which can be extracted from the slope of the T -dependent I_0 in the Arrhenius plot. The T -independent pre-factor I_{00} can be obtained from the approximation towards the inverse T equal to 0.

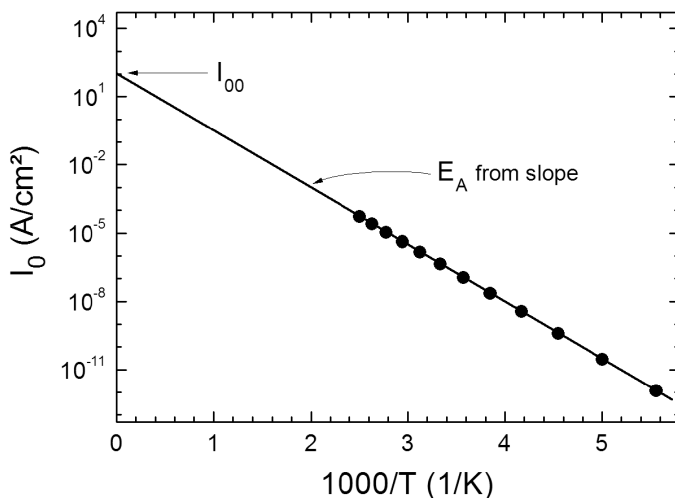


Figure 1.13. Arrhenius plot of the I_0 for a given solar cell. The logarithm of the I_0 should be multiplied with the temperature dependent ideality factor in case of a non-ideal solar cell.

The value of I_{00} amounts to about 100 A/cm^2 for the example shown in Figure 1.13.

$$I_o = I_{00} \cdot \exp\left(-\frac{E_A}{k_B \cdot T}\right) \quad (1.34)$$

The E_A is given in the unit of eV and depends on fundamental processes in the solar cell. For c-Si solar cells, the E_A is about 1.1 eV. For non-ideal solar cells, the ideality factor can also depend on T . In this case the product of the logarithm of I_0 and the T -dependent ideality factor has to be plotted in the Arrhenius plot.

The thermal activation of I_0 determines the T dependence of V_{OC} . For an ideal solar cell the T dependence of V_{OC} is given by the following equation:

$$V_{OC} = \frac{E_A}{q} - \frac{k_B \cdot T}{q} \cdot \ln\left(\frac{I_{00}}{I_{SC}}\right) \quad (1.35)$$

The V_{OC} decreases linearly with increasing T following Equation (1.35), while the slope scales with the natural logarithm of the ratio between I_{00} and I_{SC} . The maximum V_{OC} can theoretically be reached at 0 K and is equal to the E_A divided by q . Figure 1.14 shows an example for the T

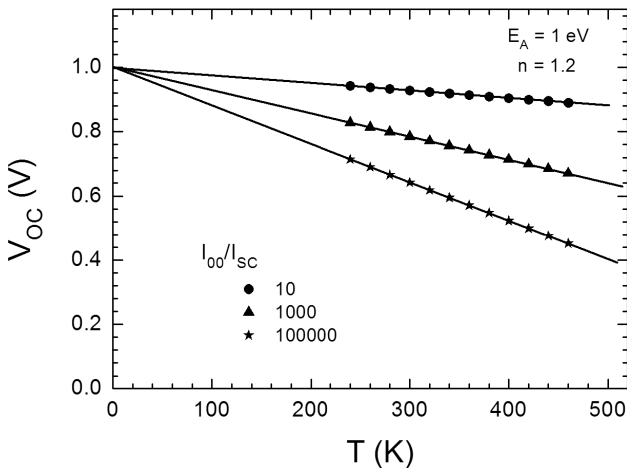


Figure 1.14. Temperature dependence of the V_{OC} for a solar cell with given activation energy and ideality factor measured at different I_{00}/I_{SC} ratios.

dependence of V_{OC} at different ratios of I_{00}/I_{SC} . As can be seen, the higher I_{SC} , the lower the T dependence of V_{OC} .

The solar energy conversion efficiency of solar cells decreases with increasing T due to the thermal activation of I_0 . Solar cells are certified for standard test conditions, i.e. at AM1.5 and 25°C. However, the T of solar cells can easily exceed 25°C and increase to values on the order of 60°C, and even higher under operation conditions (see task T1.5 in Section 1.7). This should be taken into account when calculating the expected energy production of a projected PV power plant. Further, there are solar cells with stronger or weaker T dependence of V_{OC} depending on materials, materials combinations and technology. For this reason, the temperature coefficient of PV modules is usually certified as well.

1.5.3 Measurement of I – V characteristics with loads

The measurement of complete I – V characteristics of illuminated solar cells is needed to calculate the FF and therefore the η . In addition, the R_s and R_p can be obtained from complete I – V characteristics. The ratio between the current and voltage is different at each point at an I – V characteristic, and therefore variable R_L for probing different working points are necessary. Resistors, electronic loads or capacitors (Figure 1.15) can be applied for I – V measurements depending on requirements for accuracy, costs of measurement equipment, flexibility and speed of measurement.

The use of a set of known resistances and a multimeter for voltage measurement is the easiest and cheapest way to measure an I – V characteristic of an illuminated solar cell. The resistance and the multimeter are connected in parallel with the illuminated solar cell (Figure 1.15(a)). The current is found by Ohm's law from the values of the resistance and the measured voltage.

The values of suitable R_L can be estimated by the following procedure. First, V_{OC} and I_{SC} are measured and the ratio of V_{OC} and I_{SC} , R_L^* , is calculated. The value of R_L^* is close to the R_L in the mpp. Values of resistances less than or greater than R_L^* by roughly one order of magnitude or more are chosen to get points on the I – V characteristic towards I_{SC} or towards V_{OC} , respectively, where more values of R_L have to be chosen around the mpp in order to increase the accuracy for the measurement of the FF. The described method works very well for solar cells and mini-modules

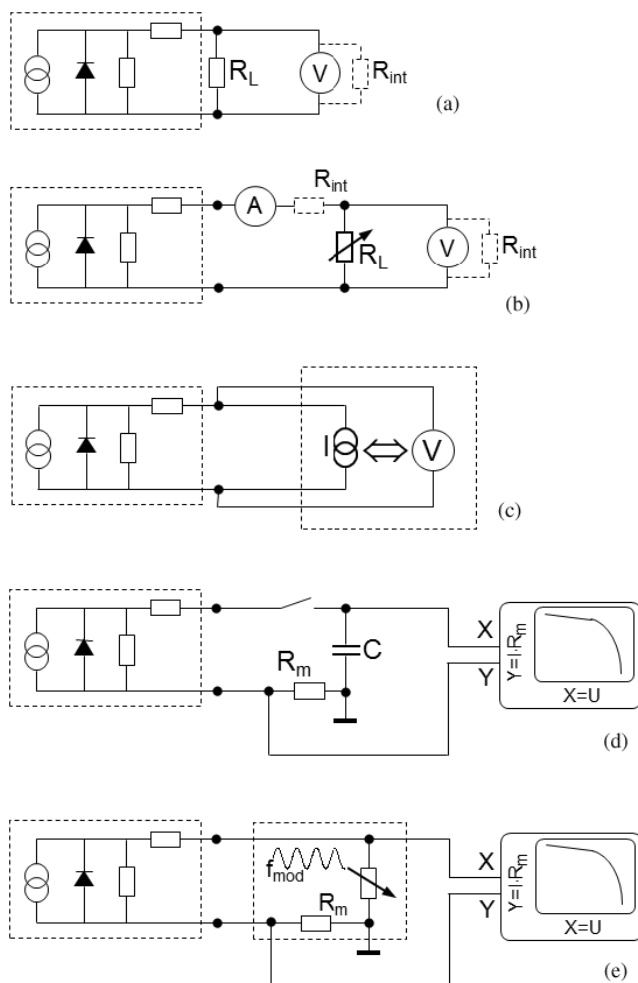


Figure 1.15. Arrangement for the measurement of I - V characteristics of solar cells for (a) a known set of resistances and a multimeter, (b) a potentiometer and two multimeters, (c) an electronic load or constant current source, (d) a switch and capacitor with oscilloscope and (e) a periodically variable R_L and oscilloscope.

with relatively low power when the R_L in the mpp is about $10\ \Omega$ or larger. The point is that contact and cable resistances can become an important source of error for low R_L . A fixed correction resistance can be introduced into the analysis if the contact resistance and the resistances of cables are constant. Furthermore, the power range of a given resistor has to be

considered since the resistance increases under heating at high power. The accuracy of the described method depends on the accuracy of the measurements of the voltage and of the resistance.

The measurement procedures of I – V characteristics and of the R_p and R_s are depicted in Figures 1.16(a)–1.16(c), respectively. The value of R_p is the negative slope of the I – V characteristic near I_{SC} where the diode current can be neglected.

$$R_p = - \left. \frac{\Delta U}{\Delta I} \right|_{U \rightarrow 0} \quad (1.36)$$

The R_s has a strong influence on the I – V characteristic towards V_{OC} . Two I – V characteristics at different intensities should be measured for

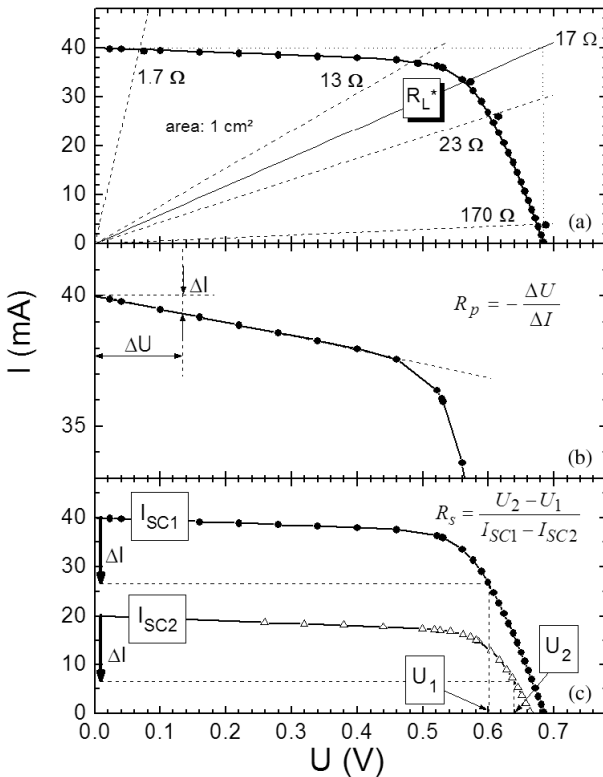


Figure 1.16. Procedure for the measurement of I – V characteristics (a) and determination of the R_p (b) and R_s (c).

obtaining R_s from a simple procedure. The two intensities have to be chosen in such a way that the respective I_{SC} (I_{SC1} and I_{SC2}) differ by a factor of roughly two. The differences between I_{SC1} and I_{SC2} and an identical current difference ΔI are determined (I_1 and I_2). It is recommended to choose a value of ΔI between about half of I_{SC1} and 90% of I_{SC2} for the measurement of R_s . The potentials U_1 and U_2 corresponding to I_1 and I_2 are found on the respective I – V characteristics (Figure 1.16(a)). The R_s is then given by

$$R_s = \frac{U_2 - U_1}{I_{SC1} - I_{SC2}} \quad (1.37)$$

The described procedure of measuring R_s works well for conventional solar cells when the FF is not very low and R_p and R_s can be well distinguished.

A potentiometer for faster variation of R_L and multimeters for current and voltage measurements can also be used for the measurement of I – V characteristics of illuminated solar cells (Figure 1.15(b)). However, one has to be careful with the current measurement since the internal resistance of the multimeter can become a series source of error, especially when changing the range of sensitivity.

A load can be simulated with a constant current source connected with the solar cell (Figure 1.15(c)). The resulting voltage drop across the solar cell is then measured. Constant current sources, also called source measure units or electronic loads, allow I – V measurements to a high degree of accuracy over a wide range of power depending on the dimensioning of the electronic load. Constant current sources can be fabricated at excellent precision so that currents down to the range of pA and less can be measured. Dark I – V characteristics and I – V characteristics of solar cells illuminated over a wide range of intensities can only be investigated with source measure units. As an aside, double-shielded cables have to be used for measurements at very low currents to avoid potential drops across long cables. Four-point probes or so-called Kelvin contacts have to be applied for measurements at high currents and/or very low R_L . In this case, the potential drop is measured between the contacts of the current input to avoid an influence of voltage drops at contact resistances for the current input and to avoid an influence of an inhomogeneous current flow.

Fast and reliable measurements of I_{SC} , V_{OC} and FF are needed for in-line control and binning of solar cells. For example, the tact cycle of a production line for c-Si solar cells is of the order of only 1 s or less. Time for handling, contacting and sorting of a solar cell is needed within one tact cycle in addition to the measurement time. This means that a c-Si solar cell has to be characterized within a time even less than 0.1–0.3 s. Very fast measurements of I – V characteristics become possible with a load changing its resistance automatically and rapidly during illumination with a light flash. The principle of such a flasher is given in Figure 1.15(d). The heart of a flasher is a capacitor (or an electronically simulated capacitor). The resistance of a capacitor varies over a huge range during charging. At the beginning of charging, the resistance of a capacitor is extremely low so that the maximum current which can be provided by the solar cell can flow. Therefore, the current flowing through the capacitor is equal to I_{SC} at the beginning of charging. At the end of charging, the resistance of the capacitor is extremely high since the charged capacitor behaves like an insulator. The maximum voltage across the charged capacitor is equal to the maximum voltage which can be provided by the solar cell, i.e. V_{OC} . The resistance of the capacitor increases continuously during the charging process so that the exact and complete I – V characteristic of the solar cell can be monitored with an oscilloscope. In the simplest case, the voltage drop across the capacitor is applied to the input of the x -channel of the oscilloscope and the current is converted with a low measurement resistance to a voltage signal and applied to the y -channel of the same oscilloscope. It is worth noting a switch connecting and disconnecting the solar cell can be applied instead of using a light pulse for flash characterization.

Rapid but not very fast measurements of I – V characteristics at high accuracy are required when an external parameter such as light intensity is continuously changed during the measurement of I – V characteristics. For such purposes the channel resistance of a field-effect transistor can be varied periodically with a frequency generator at a moderate frequency (Figure 1.15(e)). The I – V characteristic is visualized with an oscilloscope. The voltage drop across the periodically variable load is applied to the x -channel of the oscilloscope and the current transformed to a voltage signal is applied to the y -channel of the oscilloscope.

Periodically variable loads can be realized at relatively small size and low cost, and allow highly accurate and fast measurements and have an excellent performance, which makes them especially advantageous for teaching.

1.5.4 Measurement of the solar energy conversion efficiency with a pyranometer

For the measurement of the solar energy conversion efficiency of a solar cell the I - V characteristic has to be measured along with the P_{sun} at the exact moment at which the I - V -characteristic is obtained. The P_{sun} is measured with a pyranometer in the unit W/m^2 (Figure 1.17). The angles of incidence must be identical for the solar cell and for the pyranometer. The area of the solar cell has to be measured with a ruler for normalizing the power in the mpp of the solar cell to its area.

In the pyranometer, the blackbody absorbs the light of the complete sun spectrum. The decisive advantage of a blackbody is that the sunlight is absorbed over the whole relevant spectral range at the same sensitivity.

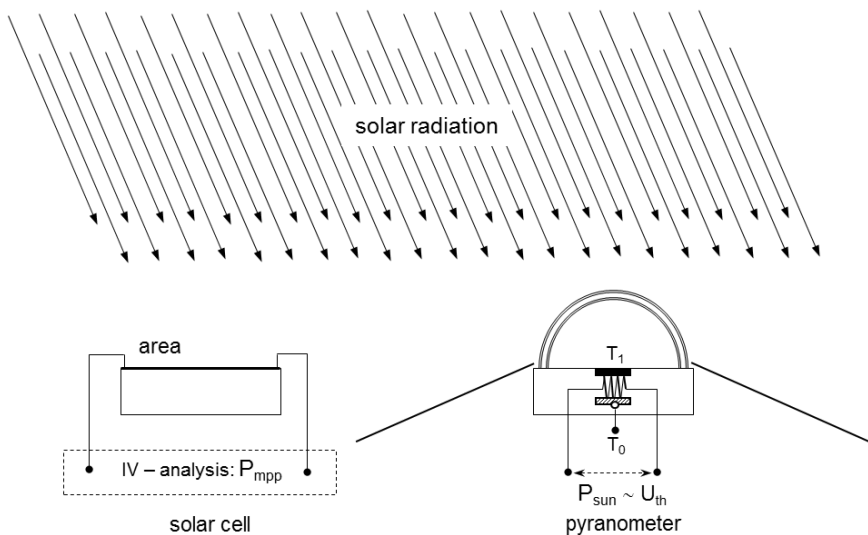


Figure 1.17. Procedure of an outdoor measurement of the solar energy conversion efficiency of a solar cell with a pyranometer.

In the blackbody, the radiation energy of the sunlight is converted into heat leading to an increase of the temperature (T_1). The T_1 is measured with a thermopile consisting of tens of thermocouples in series. A thermocouple is an intimate contact between two different metals. The contact potential between two different metals depends on the T . Therefore a thermocouple converts the T_1 into a voltage signal. The signal of the thermopile is measured as the difference of the series of contact potentials between the heated blackbody and a body at the ambient or reference temperature T_0 . The voltage signal of the thermopile at the output of the pyranometer is calibrated with respect to P_{sun} . It shall be noted that the calibration of a pyranometer demands calibration standards and is a rather critical requirement. Pyranometers of high performance have an absolute accuracy of 1% at AM1.5, i.e. 10 W/m^2 .

The blackbody in the pyranometer has to be protected from variable heat transfer, i.e. from convection of surrounding air and from changes in the heat conductivity of surrounding materials. For this purpose, the blackbody is surrounded by a double-walled glass dome and embedded into heat-insulating materials. In addition, the inner part of a pyranometer is kept dry, for example, with silica gel, in order to avoid variable heat exchange due to penetrating water molecules. Furthermore, the part of the pyranometer with the body kept at the T_0 is shaded in order to avoid changes of T_0 during the measurement.

The advantage of the measurement of η with a pyranometer is that any solar cell, PV module or string of PV modules can be characterized under given conditions at a certain moment. Therefore, accurate performance monitoring of PV power plants is coupled with real-time monitoring of P_{sun} using pyranometers.

There is a low-cost version of this which involves the measurement of P_{sun} with a photodiode. For his purpose, the dependence of I_{ph} of the photodiode on P_{sun} has to be calibrated with a pyranometer. However, in contrast to a blackbody, a silicon photodiode does not integrate over the whole relevant sun spectrum with identical sensitivity. But the sun spectrum can change depending on geography, weather conditions and time of day. Therefore, the absolute accuracy of measurements with photodiodes is in principle much lower than for measurements with a pyranometer.

1.5.5 Measurement of the solar energy conversion efficiency with a sun simulator

The measurement of η with a pyranometer has a great disadvantage which is related to the sun. The intensity and the spectrum of sunlight change dramatically over the day and depends on various conditions. This makes the direct comparison of the η of solar cells measured at different places and at different times problematic. A sun simulator overcomes the problem of comparability of measurements. The standard test conditions (STC; AM1.5 with 1000 W/m^2 and T of the solar cell 25°C) are the common standard for the characterization of the η of solar cells and PV modules (IEC, 2008).

A sun simulator is an artificial light source with an intensity spectrum very close to that of the sun at AM1.5. The artificial reproduction of the sun spectrum at AM1.5 demands appropriate light sources, combinations of light sources and filtering of light. For example, the sun spectrum at AM0 can be approximated with a blackbody with a T of 5800 K. Related blackbodies are not available on earth.

A halogen lamp can be described as a blackbody with a T of about 3000 K (Figure 1.18). The maximum intensity of a halogen lamp is at a wavelength of about 1000 nm. The simulation of the sun spectrum with

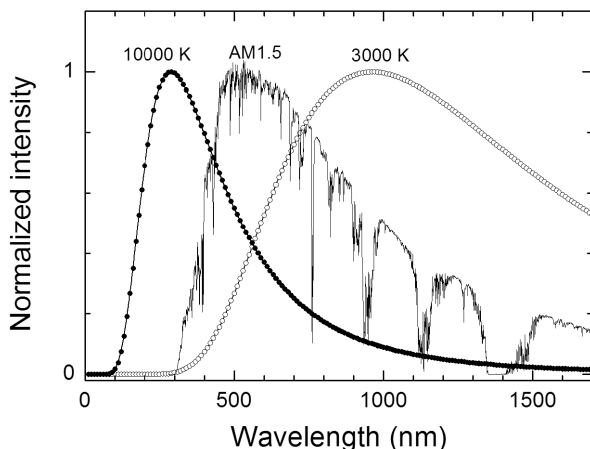


Figure 1.18. Normalized spectra of the intensity of a blackbody at 3000 K and at 10,000 K in comparison with the normalized AM1.5 spectrum.

a halogen lamp is impossible due to the very low intensity in the spectral range between blue and ultra violet light.

A xenon arc lamp is a light source with a T in the arc of about 10,000 K which is much higher than the T_S . The maximum intensity of a xenon arc lamp is at a wavelength of about 300 nm. In addition, a xenon arc lamp contains numerous spectral lines of high intensity, especially in the near infrared range. The simulation of the sun spectrum with a xenon arc lamp is nearly impossible due to the low intensity in the near infrared range.

The sun spectrum can be simulated with a combination of a halogen lamp providing enough intensity in the infrared and near infrared range and a xenon arc lamp providing high intensity in the blue and violet spectral range. Figure 1.19 shows the principle set-up of a sun simulator.

The light of the halogen and xenon arc lamps passes through a half mirror to a plane mirror from where the light is reflected to the collimator. The light passes the collimator and illuminates the solar cell. There are additional filters in the sun simulator to get rid of the intense infrared lines of the xenon arc lamp and to absorb light in the regions of decreased

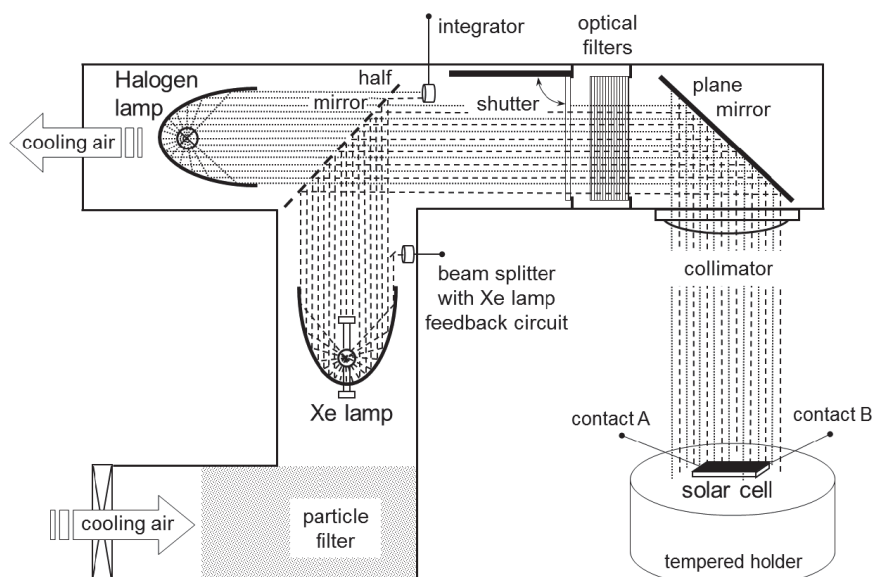


Figure 1.19. Principle set-up of a sun simulator.

intensity in the AM1.5 spectrum. In the simplest case, water is used for filtering since a significant part of absorption in the earth's atmosphere is caused by water molecules. Furthermore, a shutter is implemented to enable measurements of I – V characteristics in the dark. A beam splitter with a xenon lamp feedback circuit is needed to adjust the lamp current for keeping a constant and highly stabilized intensity of the xenon arc lamp. The integrator gives information about accumulated irradiation time. The solar cell is kept at a tempered holder at 25°C and contacted with a source measure unit. The contact resistance is usually tested with a second contact at the anode and cathode of the solar cell before the I – V characteristics are measured.

Powerful light sources of the order of a kW are needed in a sun simulator for homogeneous illumination of solar cells with an area of $10 \times 10 \text{ cm}^2$ and larger. The light sources should be actively cooled with filtered air.

The life of lamps is limited. Therefore, a sun simulator needs regular service. For example, a halogen lamp has to be replaced after every 50 hours of operation. In addition, the sun simulator has to be calibrated regularly with calibrated solar cells that are sensitive in different spectral ranges.

1.5.6 Spectral dependence of the quantum efficiency

Sunlight is white, i.e. it contains light of a wide range of wavelengths as mentioned above. The sensitivity of a solar cell strongly depends on the wavelength (λ) of the exciting light. This means that light at an equal intensity but at different λ can generate very different photocurrents. The so-called spectral response (SR) describes this property. The SR is defined as the ratio between the photocurrent measured at a given λ ($I_{\text{ph}}(\lambda)$) and the corresponding light intensity ($P_{\text{light}}(\lambda)$).

$$\text{SR}(\lambda) = \frac{I_{\text{ph}}(\lambda)}{P_{\text{light}}(\lambda)} \quad (1.38)$$

A simplified setup for the measurement of the SR is shown in Figure 1.20. The heart of this setup is a monochromator which filters required wavelengths from the spectrum of a lamp with a dispersive element such as a grating. The incoming white light is modulated with an optical chopper and the photocurrent is measured with a lock-in amplifier.

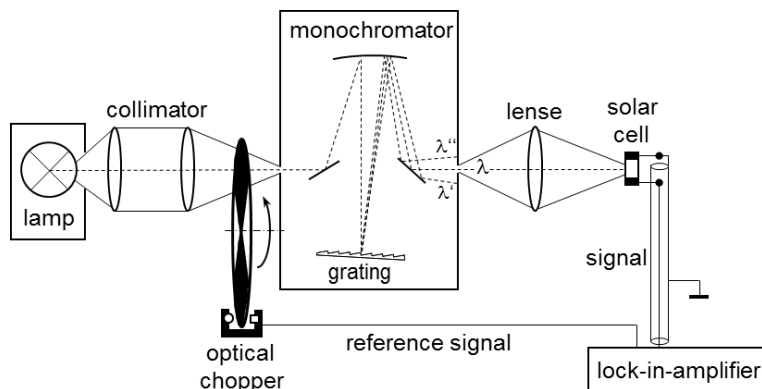


Figure 1.20. Schematic of the simplest setup for spectral response measurements.

The lock-in amplifier registers only that part of the photocurrent which is caused by the modulated incoming light, i.e. noise signals and the sensitivity to bias light are strongly suppressed. The signal at the lock-in amplifier is measured as a function of λ .

The accurate absolute calibration of a monochromatic light source demands tremendous efforts. Therefore, it is much easier to compare a measured photocurrent spectrum of an unknown solar cell ($I_{ph,1}$) with that ($I_{ph,2}$) of a solar cell which spectral response spectrum is known (SR_2). The unknown spectral response (SR_1) is found from the following equation:

$$SR_1(\lambda) = \frac{I_{ph,1}(\lambda)}{I_{ph,2}(\lambda)} \cdot SR_2(\lambda) \quad (1.39)$$

A photocurrent is the photogenerated charge per time unit and the power is the energy per time unit. The photogenerated charge is the product of the number of collected photogenerated electrons ($N_e(\lambda)$) and the q and the energy of light is the product of the number of photons ($N_{ph}(\lambda)$) and the E_{ph} . Therefore, Equation (1.38) can be transformed to

$$SR(\lambda) = \frac{q \cdot \lambda}{h \cdot c} \cdot \frac{N_e(\lambda)}{N_{ph}(\lambda)} = SR_{\max}(\lambda) \cdot \frac{N_e(\lambda)}{N_{ph}(\lambda)} \quad (1.40)$$

The maximum spectral response is proportional to the wavelength and amounts, for example, to 0.724 A/W at a λ of 900 nm.

The ratio between the number of collected photogenerated electrons and of incident photons is called quantum efficiency (QE) of a solar cell.

$$QE(\lambda) \equiv \frac{N_e(\lambda)}{N_{ph}(\lambda)} \quad (1.41)$$

Figure 1.21 shows an example of a QE spectrum. For the given example, the QE is less than one over the whole spectral range and even zero at λ below 220 nm and above 1200 nm. The QE allows for obtaining information about optical and collection losses in solar cells, which will be a topic of consideration in Chapter 2. SR and QE measurements are important for research and development of solar cells.

The I_{SC} density of a solar cell can be calculated if the SR spectrum or the QE spectrum of a solar cell is known. For this purpose, the product of the SR spectrum and the intensity spectrum of the sun or the product of the QE spectrum and the photon flux spectrum ($\Phi_{sun}(\lambda)$) of the sun are integrated over the wavelength.

$$I_{SC} = q \cdot \int_0^{\infty} SR(\lambda) \cdot P_{sun}(\lambda) \cdot d\lambda \quad (1.42')$$

$$I_{SC} = q \cdot \int_0^{\infty} QE(\lambda) \cdot \Phi_{sun}(\lambda) \cdot d\lambda \quad (1.42'')$$

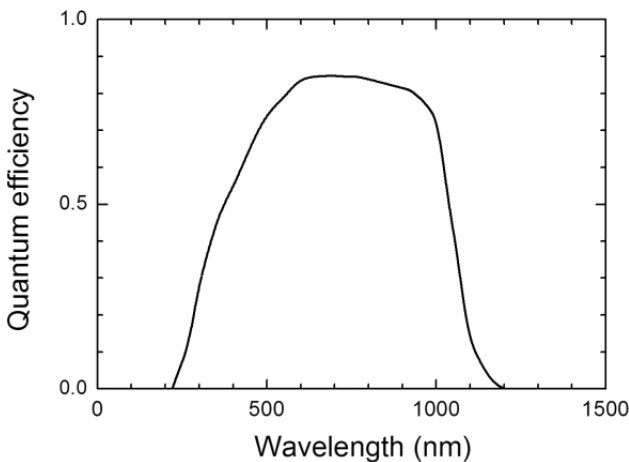


Figure 1.21. Example for a quantum efficiency spectrum of a solar cell.

The values of the I_{SC} density obtained from measurements with a sun simulator and obtained from the QE spectrum should be equal if the sun simulator and the reference solar cell for the analysis of the SR are well calibrated. Therefore, SR or QE measurements allow for an independent correct determination of I_{SC} .

1.6 Summary

The sun emits a huge amount of power or energy flux to earth characterized by the solar constant (1356 W/m^2) and by its spectral distribution with a maximum at photon energy around 1.4 eV. Total P_{sun} reaching the earth is about $1.3 \times 10^8 \text{ GW}$. For purposes of comparison, a nuclear power plant has a power of about 1 GW and a person needs on average about 0.1 kW to sustain his or her biological life. The total power received on earth can be compared with the population of mankind (more than $7 \cdot 10^9$) and the energy demand per capita for comfortable life (2 kW per capita seems sufficient). The P_{sun} reaching the earth exceeds the energy demand of mankind by several thousand times. PV solar energy conversion is aimed at using a part of this energy for human needs in form of electricity. The solar energy conversion efficiency is the most important parameter of solar cells and PV power plants. A high η combined with a low degradation rate of η and a low energy payback time is decisive for sustainable solar energy conversion.

A solar cell has two basic functions: the generation of a photocurrent and the generation of a photovoltage for the production electric power. The maximum electric power of an illuminated solar cell is the product of the I_{SC} , the V_{OC} and the FF. Loads are used to extract the power from solar cells. From an illuminated solar cell, the maximum power is extracted with a load, the resistance of which has to be equal to the quotient of the potential and of the current in the mpp. The mpp of a solar cell changes with changing illumination. This has practical consequences such as mpp tracking. For worldwide comparison, solar cells and PV modules have to be characterized at STC (power at AM1.5 with 1000 W/m^2 and temperature of the solar cell 25°C).

The behavior of solar cells can be analyzed with equivalent circuits. An ideal solar cell contains only a photocurrent generator and a diode. The

diode is necessary for internal charge separation, i.e. for the generation of a photovoltage. Ideal solar cells can be completely described by two fundamental parameters, the I_{SC} and the I_0 , i.e. all properties of materials and combinations of materials used in a solar cell are confined in I_{SC} and I_0 (Figure 1.22).

A major issue for PVs is to determine dependencies of I_{SC} and I_0 on limiting parameters related to materials and combinations of materials. The V_{OC} is a derived parameter and increases with increasing I_{SC} and/or decreasing I_0 . The FF has to be obtained from the analysis of the I - V and power-voltage characteristics. The η of ideal solar cells increases with increasing I_{SC} , i.e. under concentrated sunlight (see also Figure 1.23).

Losses caused by resistive heating at R_s and R_p have to be considered for real solar cells under operation. A $R_{s,tol}$ of solar cells with high η should be equal or less than the quotient of V_{OC} and I_{SC} of the respective ideal solar cell divided by 100. A $R_{p,tol}$ of solar cells with high η should be equal or larger than the quotient of V_{OC} and I_{SC} of the respective ideal solar cell multiplied with 100. These $R_{s,tol}$ and $R_{p,tol}$ help to evaluate the potential of given materials and combinations of materials for reaching very high η at

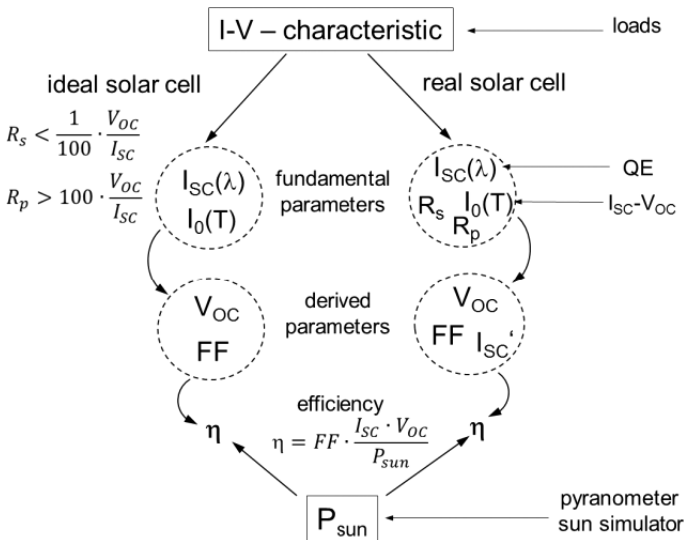


Figure 1.22. Summary of the basic characteristics of ideal and real solar cells.

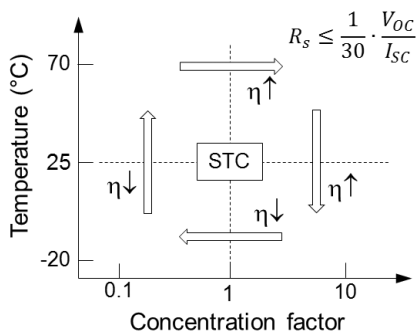


Figure 1.23. General dependence of the solar energy conversion efficiency on increasing and decreasing temperature and concentration factor of illumination.

certain operation conditions of solar cells and PV modules. For example, $R_{s,tol}$ are important economic factors for optimization of the amount of metals such as silver in bus bars on c-Si solar cells (see Chapter 7) and connecting cables in PV power plants.

At concentrated sunlight, the R_s limits FF and I_{SC} , whereas the R_p limits FF and V_{OC} at low light intensities. The η starts to reduce with increasing I_{SC} when R_s becomes larger than about a third of $R_{s,tol}$ (see also Figure 1.23). The minimum FF is 0.25. As examples, very low R_s can be realized in GaAs solar cells (see Chapter 8) and very high R_p are achieved in dye-sensitized solar cells (see Chapter 10).

The I_0 and the R_p of a solar cell can be easily obtained from I_{SC} – V_{OC} measurements. The E_A of I_0 can be derived from the dependence of the logarithm of I_0 on the reciprocal T . The thermal activation of the I_0 has the consequence that V_{OC} decreases with increasing T , i.e. η decreases with increasing T (see also Figure 1.23).

Load resistances are important for reliable measurements of I – V characteristics and therefore for the measurement of FF and R_s a solar cell. An I – V characteristic can be constructed from various working points measured precisely with different R_L . Electronic loads or constant current sources are widely used for I – V measurements. Fast I – V measurements are possible by measuring the current and the voltage during charging of a capacitor (flash operation) or by periodically applying variable R_L .

The P_{sun} can be precisely measured with a pyranometer. Standard test conditions are realized in laboratories with sun simulators. The I_{SC} densities

can be calculated by using the spectrum of the photon flux from the sun and the spectrum of the quantum efficiency of a solar cell. For the same solar cell, the I_{SC} density measured with a sun simulator should be equal to the I_{SC} density calculated from the quantum efficiency.

1.7 Tasks

T1.1: Power installed and energy produced

Ascertain the maximum energy produced by a PV power plant of 1 MW_p installed in the south of Spain compared with a PV power plant installed in Germany during the first year of operation.

T1.2: Degradation time of PV modules

Degradation rates of η of PV modules (k_{deg}) are between about 0.3–0.7%/a (c-Si) and about 0.8–1.5%/a (thin-film) (Jordan *et al.*, 2016). Plot the dependence of the degradation time of PV modules on k_{deg} after which η reduced to 90%, 80% and 50% of the initial value. Comment on the guarantee time for PV modules (usually 25 years) and on building integration of PV modules.

T1.3: Energy payback factor of PV modules

The energy payback time of PV modules (t_{EPB}) is often on the order of 1–2 years. Obtain an expression for the dependence of the energy payback factor of PV modules (EPBF) on k_{deg} and t_{EPB} . Plot EPBF for k_{deg} and t_{EPB} equal to 0.3 and 1.0%/a and to 0.5, 1 and 2 a, respectively, for a time range of up to 1000 years in logarithmic scales. Discuss the role of t_{EPB} and possible limits of EPBF.

T1.4: Photon energy in the maximum of blackbody radiation at T_s

Calculate the photon energy at which the blackbody radiation at 5800 K has a maximum.

T1.5: Wavelength of light in the maximum of blackbody radiation at T_s

Calculate the wavelength of light at which the blackbody radiation at 5800 K has a maximum.

T1.6: Maximum temperature of a solar cell under operation

Calculate the maximum temperature of solar cells with η of 10%, 20% and 40% under illumination at 1 kW/m^2 on earth when convection cooling is absent compared to solar cells on a satellite.

T1.7: Air mass

Ascertain the dependence of the air mass on the angle of incidence of sunlight and discuss the validity of the expression for large angles.

T1.8: Tolerable series and parallel resistances

Estimate the $R_{s,\text{tol}}$ and $R_{p,\text{tol}}$ for a solar cell with an I_0 of 10^{-19} A/cm^2 operated at an I_{SC} density of 25 mA/cm^2 (AM1.5) and operated at concentrated sunlight with a concentration factor of 400.

T1.9: Current–voltage characteristics

Ascertain the V_{OC} and the FFs for solar cells with an I_0 of 10^{-18} A/cm^2 , I_{SC} densities of 10 or 100 mA/cm^2 and R_p of 1 or $100 \text{ k}\Omega\text{cm}^2$.